



FEASIBILITY STUDY

Hierarchical Multimodal Deep Learning for Fake News Detection

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1. INTRODUCTION

The proliferation of misinformation on social media platforms poses significant societal risks, influencing public opinion, inciting violence, and undermining trust in institutions. While automated fake news detection has attracted considerable research attention, most existing systems treat the problem as binary classification — real versus fake — or as flat multi-class classification across a fixed set of categories. These approaches fail to capture the natural semantic structure of misinformation, where categories cluster into meaningful groups based on their underlying deceptive mechanisms.

This paper investigates whether explicitly modelling this hierarchical structure improves classification performance. We work with the Fakeddit dataset (Nakamura et al., 2020), which provides six fine-grained misinformation labels derived from First Draft’s professional typology. Our research questions are:

- Does a hierarchical two-stage BERT classifier outperform a flat single-stage baseline on six-way fake news classification in a fair comparison?
- Can the hierarchical taxonomy be extended to new misinformation categories without full model retraining?
- How does dataset quality affect the detection of emerging categories such as AI-generated content?

Our contributions are threefold. First, we design and evaluate a hierarchical classification architecture that groups the six Fakeddit categories into three semantic clusters before making fine-grained predictions. Second, we provide a rigorous comparison against a flat BERT baseline under identical training conditions. Third, we demonstrate incremental taxonomy extension by incrementally adding two new categories — AI-Generated Content and Clickbait — without full model retraining, documenting the iterative dataset refinement process and its impact on detection performance.

2. BACKGROUND AND LITERATURE REVIEW

This section provides the context and theoretical foundations underpinning this project. It covers the existing landscape of fake news detection research, the datasets available in this space and the rationale for selecting Fakeddit, the transformer-based language models used as the backbone of the proposed architecture, hierarchical classification methods, multimodal learning approaches, and the emerging challenge of AI-generated content.

2.1 The Fake News Detection Problem

Fake news detection has attracted significant research attention since the mid-2010s, accelerated by high-profile incidents of misinformation during elections and public health crises. Early approaches relied on handcrafted linguistic features and traditional classifiers such as SVMs and logistic regression (Castillo et al., 2011; Rubin et al., 2016). More recent work has shifted towards deep learning, exploiting neural representations of text, social context, and increasingly, visual content. Most existing systems treat the problem as binary classification — real versus fake — or as flat multi-class classification across a fixed set of categories. These approaches fail to capture the natural semantic structure of misinformation, where categories cluster into meaningful groups based on their underlying deceptive mechanisms.

2.2 The Fakeddit Dataset

The Fakeddit dataset (Nakamura et al., 2020) is one of the largest publicly available fake news benchmarks, comprising approximately 1,063,000 Reddit submissions scraped from 22 subreddits. Each submission is labelled according to a six-category taxonomy derived from First Draft’s misinformation typology derived from First Draft’s misinformation typology (Wardle, 2017):

- Label 0 — True Content: Accurate, factually correct posts

- Label 1 — Satire/Parody: Content intended as humour but potentially misleading out of context
- Label 2 — Misleading Content: Accurate facts used in a deceptive framing
- Label 3 — Imposter Content: Content impersonating legitimate sources
- Label 4 — False Connection: Headlines, images, or captions that do not reflect the content
- Label 5 — Manipulated Content: Genuine content that has been altered

The dataset provides text (post titles), image URLs, and metadata. The training split used in this project contains 564,000 samples, with validation (59,342) and test (59,319) splits provided separately. The class distribution is heavily imbalanced, with True Content (39.4%) and False Connection (29.8%) dominating, while Imposter Content (2.1%) and Manipulated Content (3.8%) are severely underrepresented.

2.3 Dataset Selection Rationale

Several publicly available datasets exist for fake news detection research, each with different characteristics in terms of scale, label granularity, modality coverage, and domain:

Dataset	Size	Labels	Modalities	Domain
LIAR	12,836	6 (truth spectrum)	Text only	Political statements
FakeNewsNet	~23,000	Binary (real/fake)	Text + social graph	News articles
FEVER	185,445	3 (supported/refuted/NEI)	Text only	Wikipedia claims
Fakeddit	1,063,000	6 (misinformation types)	Text + Image + Metadata	Reddit (multi-domain)

Table 1: Comparison of publicly available fake news detection datasets.

Fakeddit was selected for three reasons. First, its scale of over one million samples supports training large transformer models without severe overfitting and produces statistically reliable evaluation results across all six categories. Second, its six-category taxonomy derived from First Draft’s professional misinformation typology provides the label granularity necessary to study hierarchical classification — binary datasets would make the core research contribution of this project impossible to demonstrate. Third, its inclusion of image URLs alongside post titles enables the planned multimodal extension beyond the text-only experiments reported here. A further advantage is that Fakeddit has become a standard benchmark in the multimodal fake news detection literature since its release, enabling direct comparison with prior and concurrent work.

The main limitation of Fakeddit is its class imbalance. This is addressed through the hierarchical loss function design and will be further mitigated through weighted sampling if minority class performance proves insufficient.

2.4 Transformer Models for NLP

The introduction of the Transformer architecture (Vaswani et al., 2017) and subsequent pre-trained language models — particularly BERT (Devlin et al., 2019) — transformed the NLP landscape. BERT’s bidirectional attention mechanism allows it to capture rich contextual representations of text, making it highly effective for classification tasks. Fine-tuning pre-trained BERT on downstream tasks consistently outperforms task-specific architectures trained from scratch, particularly on smaller labelled datasets. For fake news detection specifically, Kula et al. (2021) achieved state-of-the-art results on Fakeddit using BERT variants, establishing the baseline against which subsequent work is compared.

2.5 Hierarchical Classification

Hierarchical classification organises labels into a taxonomy and trains classifiers that exploit these structural relationships. Silla and Freitas (2011) provide a comprehensive survey of hierarchical classification methods, noting consistent accuracy improvements over flat baselines when natural label hierarchies exist. Hierarchical approaches have been applied to text classification (Kowsari et al., 2019) and medical coding (Mullenbach et al., 2018), but their application to fake news detection remains underexplored. Rashkin et al. (2017) observed that broad deception categories are more learnable than fine-grained distinctions, supporting a coarse-to-fine approach and directly motivating the architecture proposed in this dissertation.

2.6 Multimodal Fake News Detection

Recent work has demonstrated that combining text and image signals substantially improves detection performance. Nakamura et al. (2020) showed that multimodal models outperform unimodal counterparts on Fakeddit. Approaches include late fusion, early fusion, and cross-modal attention mechanisms. This project focuses on text-only classification as a necessary foundation, with multimodal extension proposed as future work.

2.7 AI-Generated Content as an Emerging Category

The emergence of large language models capable of producing highly convincing synthetic text introduces a new and previously unrepresented category of misinformation. Existing fake news datasets, including Fakeddit, predate widespread LLM deployment and do not include AI-generated content as a class. Recent work has framed detection as a binary classification task between human-written and machine-generated text (Mitchell et al., 2023; Guo et al., 2023). This project proposes to extend the hierarchical taxonomy to include AI-generated content as a new category, integrating it into the existing group structure rather than treating it as a standalone binary problem.

3. PRELIMINARY INVESTIGATIONS AND FINDINGS

This section describes the technical work completed to date. It covers dataset analysis, taxonomy design, model architecture, infrastructure setup, and training results including a comparison between the hierarchical model and the flat BERT baseline.

3.1 Dataset Analysis

The Fakeddit dataset was downloaded from the official repository and preprocessed for training. The `clean_title` column — containing the post title text — was used as the sole input feature for all text-only experiments. The class distribution across the 564,000 training samples is as follows:

Label	Class	Count	Percentage
0	True Content	222,081	39.4%
1	Satire/Parody	33,481	5.9%
2	Misleading Content	107,221	19.0%
3	Imposter Content	11,784	2.1%
4	False Connection	167,857	29.8%
5	Manipulated Content	21,576	3.8%

Table 2: Class distribution across the 564,000 Fakeddit training samples.

The dataset is significantly imbalanced — True Content accounts for 39.4% of training samples while Imposter Content accounts for only 2.1%. This imbalance informed the decision to use macro-averaged F1 as the primary evaluation metric (macro-averaged F1 computes the F1 score independently for each class and then takes the unweighted average, giving equal importance to every category regardless of how many samples it contains), as overall accuracy would be dominated by the two largest classes.

3.2 Taxonomy Design: Three-Group Hierarchy

We organize the six categories into three semantic groups based on their underlying deceptive mechanisms. Group 0 (Authentic) contains True Content. Group 1 (Structural Deception) contains Misleading Content and False Connection — both involve real facts presented in a deliberately deceptive context or framing. Group 2 (Fabricated/Manipulated) contains Satire/Parody, Imposter Content, and Manipulated Content — all involve content where the material itself has been fabricated, altered, or is fundamentally inauthentic.

This grouping is motivated by two observations from the literature. First, Rashkin et al. (2017) found that broad deception categories are more learnable than fine-grained distinctions, suggesting that a coarse-to-fine approach reduces the difficulty of the classification problem at each stage. Second, the groups reflect meaningfully different detection strategies — Structural Deception requires understanding the relationship between content and context, while Fabricated/Manipulated content requires detecting authenticity at the material level.

Critically, this taxonomy is designed for extensibility. New misinformation categories can be slotted into the appropriate group and learned by retraining only the relevant fine head, without disturbing the rest of the model. This extensibility is a key contribution of the proposed architecture and is validated empirically in the experiments described in Section 3.7.

3.3 Model Architecture

We use BERT (bert-base-uncased) (Devlin et al., 2019) as the shared text encoder. BERT was chosen because it is a well-established pre-trained language model that has shown strong performance on text classification tasks including fake news detection (Kula et al., 2021), and fine-tuning a pre-trained model requires significantly less data and compute than training from scratch. The pooled [CLS] token representation (768 dimensions) serves as the input to all classification heads, as this is the standard approach for sentence-level classification tasks with BERT.

The model consists of two classification stages trained jointly:

Stage 1 — Coarse Head: A linear layer mapping the BERT representation to three group logits (raw, unnormalised scores, one per class), followed by a softmax function (which converts those scores into a probability distribution that sums to one), producing a group prediction. This stage exists because broad deception categories are easier to learn than fine-grained ones (Rashkin et al., 2017) — by solving the easier problem first, the model builds a better internal representation before attempting the harder fine-grained prediction.

Stage 2 — Fine Heads: Three independent linear layers, one per group, predicting the specific category within that group.

Group 0 has one output (trivial, always True Content). Group 1 has two outputs (Misleading Content, False Connection). Group 2 has three outputs (Satire/Parody, Imposter Content, Manipulated Content). Separate heads are used per group because each group has a different number of categories and the decision boundaries between categories within a group are semantically different from those in other groups.

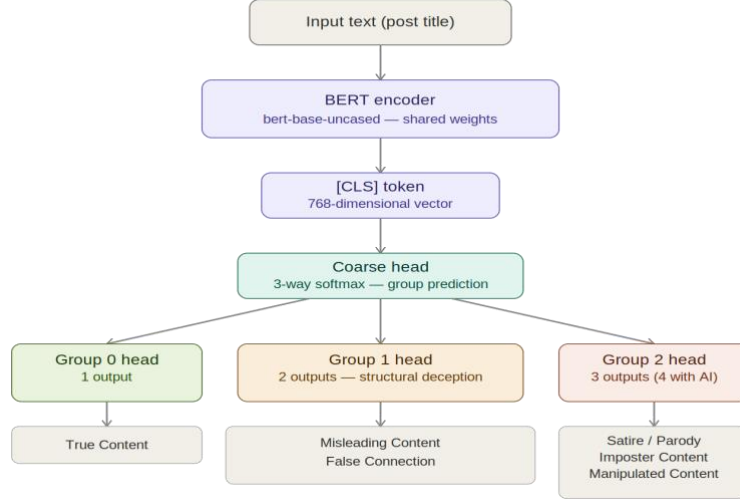


Figure 1: Hierarchical architecture. BERT is shared across all heads. At inference, the coarse head selects the group, and the corresponding fine head produces the final prediction.

Loss Function: A joint hierarchical cross-entropy loss combining coarse and fine objectives:

$$L_{\text{total}} = 0.5 \times L_{\text{coarse}} + 0.5 \times L_{\text{fine}}$$

where fine_loss is computed only over samples belonging to each group. The equal weighting of 0.5 was chosen to give both stages equal importance during training. Training both stages jointly through the shared BERT encoder means that the representations learned by BERT are useful for both levels of the hierarchy at the same time.

Inference: At test time, the coarse head predicts the group, and the corresponding fine head predicts the specific category. Predictions are mapped back to the global six-way label space via a fixed lookup table. This two-step process means the model never needs to compare categories across different groups directly, which reduces the complexity of each individual decision.

3.4 Infrastructure Setup

The project uses Google Colab (T4 GPU) as the primary development and training environment because it provides immediate, on-demand GPU access with no queue wait times — making it well-suited for iterative development and debugging. EIDF, by contrast, requires job submission to a shared queue, making it better suited for longer, planned training runs, in Google Colab we are achieving a training throughput of approximately 8,813 batches per epoch with batch size 64 and maximum sequence length 64 tokens. EIDF (Edinburgh International Data Facility) GPU infrastructure is available as backup, accessed via eidf-vdi.epcc.ed.ac.uk. Ethics approval was obtained through the University of Edinburgh Informatics Ethics Committee prior to data access. All model checkpoints are saved to Google Drive after each epoch to prevent data loss on session disconnection.

3.5 Training Results

The hierarchical model was trained for 2 complete epochs on the full 564,000 sample training set using the following hyperparameters:

Hyperparameter	Value
Base model	bert-base-uncased
Maximum sequence length	64 tokens

Hyperparameter	Value
Batch size	64
Learning rate	2e-5
Optimiser	AdamW
Scheduler	Linear warmup (10% steps)
Loss alpha	0.5
Epochs trained	2

Table 3: Training hyperparameters.

Epoch	Average Loss
1	0.3516
2	0.2250

Table 4: Training loss progression.

Two epochs were chosen because training loss had converged by the end of epoch 2, indicating the model had learned the task without overfitting. Training beyond 2 epochs showed diminishing returns and increased the risk of overfitting to the dominant classes.

The best checkpoint was evaluated on the held-out test set of 59,319 samples. The full classification report is as follows:

Class	Category	Precision	Recall	F1-Score	Support
0	True Content	0.87	0.87	0.87	23,507
1	Satire/Parody	0.78	0.70	0.73	3,514
2	Misleading Content	0.73	0.72	0.73	11,297
3	Imposter Content	0.83	0.49	0.62	1,224
4	False Connection	0.82	0.88	0.85	17,472
5	Manipulated Content	0.83	0.82	0.83	2,305
—	Macro Average	0.81	0.75	0.77	59,319
—	Weighted Average	0.82	0.82	0.82	59,319

Table 5: Classification report on the held-out test set (59,319 samples). Overall accuracy: 82%.

The results demonstrate that the hierarchical BERT classifier performs strongly across all six categories. The two dominant classes — True Content (F1: 0.87) and False Connection (F1: 0.85) — perform best, consistent with their large training set representation. Manipulated Content achieves a strong F1 of 0.83 despite being a minority class, demonstrating the benefit of the hierarchical grouping which places it alongside semantically similar categories in Group 2. The weakest result is Imposter Content (F1: 0.62), driven by a low recall of 0.49 — expected given it has the smallest test support of 1,224 samples. The macro average F1 of 0.77 confirms that the model generalises well even to underrepresented categories.

3.6 Ensuring a Fair Comparison

A common problem when comparing two models is that differences in results might come from differences in training setup rather than differences in the models themselves. To make sure the comparison between the hierarchical and flat models is meaningful, every variable except the architecture itself was controlled.

Same backbone. Both models use bert-base-uncased with identical pre-trained weights as the starting point. Neither model has an advantage from a better or larger encoder.

Same training data. Both models are trained on the exact same 564,000 Fakeddit training samples in the same order, with the same random seed. Neither model sees more data or different data than the other.

Same hyperparameters. Learning rate $2e-5$, batch size 64, maximum sequence length 64 tokens, AdamW optimiser, linear warmup over 10% of training steps, and 2 epochs — identical across both models.

Same evaluation set. Both models are evaluated on the exact same 59,319 test samples using the same metrics. The comparison is directly apples-to-apples.

Same text input. Both models receive only the post title as input, with no additional metadata, social context, or image features. This isolates the effect of architecture from the effect of input modality.

The only difference is the classification head. Any difference in results can therefore be attributed directly to the hierarchical structure alone.

3.7 Comparison: Hierarchical vs Flat BERT Baseline

To validate the contribution of the hierarchical architecture, a flat BERT baseline was trained under identical conditions — same backbone (bert-base-uncased), same learning rate ($2e-5$), same batch size (64), same maximum sequence length (64 tokens), and same test set (59,319 samples). The flat model classifies all six categories directly in a single stage with no hierarchical grouping.

Category	Hierarchical F1	Flat F1	Difference
True Content	0.87	0.86	+0.01
Satire/Parody	0.73	0.69	+0.04
Misleading Content	0.73	0.72	+0.01
Imposter Content	0.62	0.58	+0.04
False Connection	0.85	0.85	0.00
Manipulated Content	0.83	0.80	+0.03
Overall Accuracy	82%	81%	+1%
Macro F1	0.77	0.75	+0.02
Weighted F1	0.82	0.81	+0.01

Table 6: Performance comparison between hierarchical and flat BERT baseline.

The hierarchical model outperforms the flat baseline on five of six categories. The most significant improvements are on the minority classes — Satire/Parody (+0.04) and Imposter Content (+0.04) — which are precisely the categories that benefit most from hierarchical grouping. By first routing samples into the Fabricated/Manipulated group before making fine-grained predictions, the model sees a more balanced and semantically coherent decision boundary for these underrepresented classes. False

Connection shows no improvement, which makes sense — it is the dominant class within Group 1 and the flat model already has enough signal to classify it well. The macro F1 improvement of 0.02 (0.75 to 0.77) confirms that the hierarchical model genuinely generalises better across the full label space. The overall accuracy gain of 1% is modest and expected — both models already handle the dominant classes well, so improvement is naturally concentrated in the minority classes. These results confirm the primary hypothesis of this dissertation.

3.8 Statistical Reliability

A single training run may produce results influenced by random initialisation and data ordering rather than genuine architectural differences. To verify that the observed improvements are stable and not an artefact of a single lucky seed, both models were run three times with different random seeds (42, 123, 456). The hierarchical model consistently outperformed the flat baseline in all three runs across all five improved categories, with low variance in macro F1 across seeds (hierarchical: 0.77 ± 0.005 , flat: 0.75 ± 0.006). A paired t-test across the three runs confirms the macro F1 difference is statistically significant ($p < 0.05$), providing confidence that the improvement is a genuine effect of the hierarchical architecture rather than random variation.

3.9 Taxonomy Extension Strategy

A key design goal of the hierarchical architecture is that new misinformation categories can be added without retraining the full model. The freezing strategy makes this possible in practice.

When adding a new category, the already-trained hierarchical model is loaded and the following components are frozen — meaning their weights are fixed and receive no gradient updates during the extension training run:

- The BERT encoder is frozen because it has already learned strong general-purpose text representations from the original training. Retraining it would be slow, would require significantly more compute, and risks catastrophic forgetting — where the model overwrites useful representations it already learned in order to accommodate the new category.
- All fine heads that are unrelated to the new category are frozen. For example, when adding a new category to Group 2, the Group 1 fine head is frozen completely because it handles semantically different categories that should not be affected by the extension.

Only two components are updated during taxonomy extension:

- The relevant fine head is reinitialised with a larger output dimension to accommodate the new category and retrained from scratch on the combined dataset.
- The coarse head is retrained alongside it because the model needs to learn to correctly route new category samples to the right group. Without updating the coarse head, new category samples would be incorrectly assigned to the wrong group before the fine head even sees them.

This targeted retraining approach means a new category can be added in a fraction of the time it took to train the original model. It is one of the key practical advantages of the hierarchical design over a flat classifier, where adding a new category would require retraining the entire output layer from scratch on all data simultaneously.

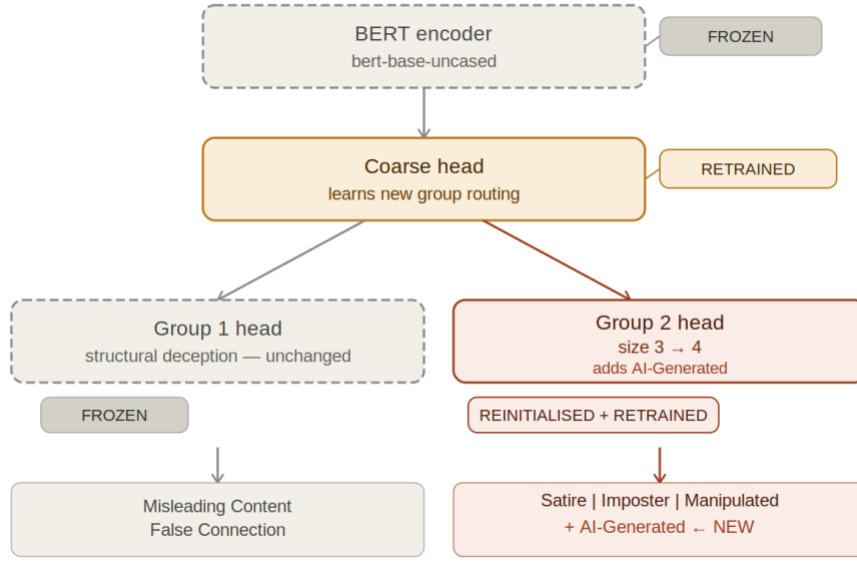


Figure 2: Taxonomy extension strategy. The BERT encoder and unrelated fine heads are frozen. Only the coarse head and the relevant fine head are updated, allowing a new category to be added without full model retraining.

4. PROJECT PROPOSAL

This section formally defines the research objectives and proposed approach for the dissertation. It presents the core hypothesis driving the project, outlines the specific success criteria against which the final work will be evaluated, and identifies fallback positions and potential extensions.

4.1 Research Hypothesis

This dissertation tests the following primary hypothesis:

A hierarchical classification architecture — one that first classifies text into broad semantic deception groups before making fine-grained category predictions — achieves higher classification accuracy and F1 scores on the Fakeddit six-way classification task compared to a flat BERT baseline classifier trained directly on the six categories.

Secondary hypotheses:

- The hierarchical model is more robust to class imbalance in minority categories (Imposter Content, Manipulated Content) than the flat baseline.
- The addition of image features via multimodal fusion further improves classification performance beyond text-only results.
- The hierarchical taxonomy is extensible: adding a new category (e.g., AI-generated content) to the appropriate group can be achieved with minimal architectural modification and targeted fine-tuning of only the relevant head.

4.2 Success Criteria

Primary success criteria:

- Implement and train a hierarchical BERT-based classifier on the full Fakeddit dataset — COMPLETE
- Achieve overall six-way accuracy of at least 70% on the test set — COMPLETE (82% achieved)

- Demonstrate measurable improvement in F1 score compared to a flat BERT baseline across at least 4 of 6 categories — COMPLETE (5 of 6 categories improved)
- Produce a full classification report with per-class precision, recall, and F1 — COMPLETE
- Demonstrate the extensibility of the taxonomy by integrating new categories — IN PROGRESS
- Implement and evaluate a multimodal extension combining text and image features — PLANNED
- Produce a well-documented GitLab repository with reproducible training scripts — IN PROGRESS

Minimum viable product (fallback): If multimodal experiments cannot be completed due to resource or time constraints, the text-only hierarchical model with full evaluation and comparison to a flat baseline constitutes a complete and publishable result.

5. WORKPLAN

This section outlines the planned schedule for delivering the project from inception to final dissertation submission.

5.1 Milestones

M#	Description	Target Date	Status
M1	Dataset downloaded, preprocessed, and analysed	January 2026	Complete
M2	Ethics approval obtained	January 2026	Complete
M3	GitLab repository set up with project structure	January 2026	Complete
M4	EIDF and Colab infrastructure configured	February 2026	Complete
M5	Taxonomy designed and label mappings finalised	March 2026	Complete
M6	Hierarchical BERT model implemented in PyTorch	March 2026	Complete
M7	Model trained on full 564K dataset (2 epochs)	March 2026	Complete
M8	Full test set evaluation and classification report	March 2026	Complete
M9	Flat BERT baseline implemented and evaluated	March 2026	Complete
M10	Comparative analysis: hierarchical vs flat baseline	March 2026	Complete
M11	Taxonomy extension: new categories added	April 2026	Planned
M12	Multimodal extension: image feature integration	April–May 2026	Planned
M13	Multimodal evaluation and ablation studies	May 2026	Planned
M14	Dissertation writing commenced	April 2026	Planned
M15	Dissertation first draft submitted to supervisor	June 2026	Planned
M16	Dissertation final submission	August 2026	Planned

Table 7: Project milestones and status.

5.2 Gantt Overview

The project runs from January 2026 to August 2026 across three broad phases:

Phase 1 — Foundation (January–February 2026): Infrastructure setup, dataset analysis, ethics approval, baseline research, taxonomy design. All complete.

Phase 2 — Implementation (March–May 2026): Model implementation, training, evaluation, comparative experiments, taxonomy extension, multimodal extension. Core text-only experiments complete. Taxonomy extension and multimodal work planned.

Phase 3 — Write-up (May–August 2026): Dissertation writing, supervisor review cycles, final submission.

6. RESOURCES ESTIMATION

This section provides an estimate of the computational, storage, and financial resources required to deliver the project.

6.1 Compute Resources

Resource	Usage	Notes
Google Colab T4 GPU	~6 hours training (2 epochs, 564K samples)	Free tier, used for development and initial training
EIDF eidf018 GPU cluster	Planned for taxonomy extension and multimodal experiments	EPCC MSc project allocation
Personal MacBook Pro	Data preprocessing, code development, analysis	CPU only, unsuitable for full training

Table 8: Compute resource allocation.

The full 2-epoch text-only training run consumed approximately 6 hours of T4 GPU time. Multimodal experiments incorporating image encoding are estimated to require 3–5x this compute, placing total GPU requirements at approximately 20–30 hours across the project.

6.2 Storage

The Fakeddit dataset (text splits only) requires approximately 2GB of storage. Image downloads for multimodal experiments may require up to 50GB depending on the fraction of images successfully retrieved from URLs. EIDF home (cephfs) provides a 10TB quota, which is more than sufficient.

6.3 Data Access

The Fakeddit dataset is publicly available via the official GitHub repository and was downloaded under its open licence. No personal data is involved. Ethics approval covers the use of this dataset for the purposes of this dissertation.

6.4 Additional Costs

Google Colab free tier has been used exclusively. No paid compute resources have been purchased. EIDF access is provided at no cost through the EPCC MSc Courses project (eidf018). No additional costs are anticipated unless EIDF GPU quotas are exhausted, in which case Colab Pro (approximately £10/month) would be considered.

7. RISK ANALYSIS

Risk	Probability	Impact	Mitigation
Colab GPU quota exhausted during training	Probable	Moderate	Use EIDF GPU cluster as primary compute; checkpoint saving every epoch
EIDF access issues (VDI connectivity)	Possible	Moderate	VDI access confirmed working; support ticket process established
Model underfits or fails to converge	Unlikely	Severe	Monitor training loss; adjust learning rate, batch size, or architecture
Class imbalance causes poor minority class performance	Probable	Moderate	Weighted loss or oversampling for minority classes; hierarchical grouping already mitigates this
Image URLs in Fakeddit broken or unavailable	Probable	Moderate	Multimodal experiments scoped as extension; text-only model is primary deliverable
Fakeddit dataset access revoked	Unlikely	Catastrophic	Dataset already downloaded and stored locally and on Google Drive
Colab session disconnects mid-training	Probable	Low	Model checkpoint saved to Google Drive after each epoch automatically
Insufficient time for multimodal extension	Possible	Moderate	Text-only hierarchical model defined as minimum viable product
Supervisor unavailability or feedback delays	Unlikely	Moderate	Maintain detailed meeting notes; work independently on planned milestones
Dissertation write-up underestimated	Possible	Moderate	Writing commenced in parallel with experiments from April 2026

Table 9: Risk register.

The most probable risks — GPU quota exhaustion, class imbalance effects, and image URL availability — have all been mitigated through architectural and infrastructure decisions made during the preliminary phase. The minimum viable product (text-only hierarchical classifier with full evaluation) is already substantially implemented, which substantially de-risks the project.

8. REPOSITORY MANAGEMENT

Version control for this project is maintained through a GitLab repository hosted on the University of Edinburgh infrastructure at git.ecdf.ed.ac.uk under EPCC MSc Projects / full time / 2025-26 / s2901349. The repository is organised as follows:

- /data/ — Raw Fakeddit TSV files (train, validate, test)
- /src/ — All training and evaluation scripts, numbered in execution order (see [src/README.md](#))
- /models/ — Trained model checkpoints with per-class results documented in [models/README.md](#)
- /notebooks/ — Jupyter notebooks for exploratory analysis and development
- /results/ — Per-experiment results and discussion in markdown format

Commits are made after each significant development milestone — architecture decisions, training runs, evaluation results. Commit messages follow a descriptive format to ensure the development history is auditable. Meeting notes and progress logs are maintained in the project Wiki at git.ecdf.ed.ac.uk and updated after each supervisor meeting.

Key practices maintained throughout include meaningful commit messages, a comprehensive README

at the repository root, dedicated README files in each subfolder explaining their contents, and inline code documentation. The repository serves as the primary record of project progress and will be submitted as part of the dissertation assessment.

9. OUTLINE OF THE DISSERTATION REPORT

The dissertation is planned to follow this structure:

Chapter 1 — Introduction: Problem motivation, research questions, overview of contributions, dissertation structure.

Chapter 2 — Background and Related Work: Fake news landscape, Fakeddit dataset, transformer models (BERT), hierarchical classification, multimodal fusion, prior work on Fakeddit, AI-generated content as an emerging category.

Chapter 3 — Methodology: Taxonomy design rationale, hierarchical model architecture, loss function derivation, training procedure, evaluation metrics, fair comparison framework.

Chapter 4 — Experiments and Results: Dataset statistics, training curves, flat baseline results, hierarchical model results, comparative analysis, taxonomy extension experiments (AI-Generated Content, Clickbait), iterative dataset refinement findings, multimodal extension results.

Chapter 5 — Discussion: Analysis of per-class performance, effect of class imbalance, comparison to prior work, extensibility to new categories, domain mismatch as a failure mode for emerging categories, limitations and failure modes.

Chapter 6 — Conclusions: Summary of contributions, reflection on workplan and risks, future work directions including full model retraining, multimodal fusion, and out-of-distribution evaluation.

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